*This is your first homework assignment. I strongly encourage you to typeset it using the LaTeX template provided. It is due by ~~Monday, January 13th.~~ Wednesday, January 22nd.*

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(1) Verify that the following matrices are orthogonal. *Note there may be a little bit of rounding error.*

$$Q_1 = \begin{bmatrix} 0.1817 & 0.7236 & 0.6658 \\ 0.6198 & -0.61 & 0.4938 \\ -0.7634 & -0.3230 & 0.5594 \end{bmatrix} \quad Q_2 = \begin{bmatrix} 0.6404 & 0.3577 & -0.6797 \\ -0.3121 & 0.9298 & 0.1952 \\ 0.7010 & 0.0872 & 0.7071 \end{bmatrix}$$

(2) Consider the least squares problem $\mathbf{x}^* = \operatorname{argmin}_{\mathbf{x}\in\mathbb{R}^4} \|A\mathbf{x} - \mathbf{b}\|_2$ where $A$ and $\mathbf{b}$ are given below. Find $\mathbf{x}^*$ using the QR decomposition method, where $Q$ and $R$ are also given below.

$$A = \begin{bmatrix} 0.7922 & 0.6787 & 0.7060 & 0.6948 \\ 0.9595 & 0.7577 & 0.0318 & 0.3171 \\ 0.6557 & 0.7431 & 0.2769 & 0.9502 \\ 0.0357 & 0.3922 & 0.0462 & 0.0344 \\ 0.8491 & 0.6555 & 0.0971 & 0.4387 \\ 0.9340 & 0.1712 & 0.8235 & 0.3816 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 0.7655 \\ 0.7952 \\ 0.1869 \\ 0.4898 \\ 0.4456 \\ 0.6463 \end{bmatrix}$$

$$Q = \begin{bmatrix} -0.4191 & -0.1861 & -0.5979 & -0.1311 & -0.2584 & -0.5901 \\ -0.5076 & -0.1321 & 0.5495 & -0.3017 & -0.5548 & 0.1553 \\ -0.3469 & -0.4172 & -0.1468 & 0.7373 & -0.0250 & 0.3739 \\ -0.0189 & -0.5331 & -0.2732 & -0.5822 & 0.3160 & 0.4494 \\ -0.4492 & -0.0950 & 0.3924 & 0.0215 & 0.7089 & -0.3637 \\ -0.4941 & 0.6933 & -0.3005 & -0.0938 & 0.1500 & 0.3919 \end{bmatrix} \quad R_1 = \begin{bmatrix} -1.8902 & -1.3134 & -0.8595 & -1.1681 \\ 0 & -0.6892 & 0.2859 & -0.3632 \\ 0 & 0 & -0.6673 & -0.3327 \\ 0 & 0 & 0 & 0.4674 \end{bmatrix}$$

(3) **(Optional)** Verify your answer from question 2 using a computer. You can use either $A \setminus \mathbf{b}$ in MATLAB or `np.linalg.lstsq(A,b)` in Python. Note that you can also verify the QR decomposition by using the command `qr(A)` in MATLAB or `np.linalg.qr(A)` in Python.

(4) Prove that if $Q_1$ and $Q_2$ are orthogonal then so is $Q_1Q_2$. *Hint: There are multiple ways to show this. You may want to prove that all the columns of the new matrix $Q_1Q_2$ are orthonormal. Or you may want to show that $(Q_1Q_2)^{-1} = (Q_1Q_2)^\top$.*

(5) For all parts of this question let $A = \begin{bmatrix} 3 & 3 & 1 \\ 3 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix}$

(a) Find the eigenvalues ~~and corresponding eigenvectors of $A$.~~

(b) Use your answer to part (a) to compute $\|A\|_2$.

(c) **(Optional)** Verify your answer to part (a) using a computer. You can use either `eig(A)` on MATLAB or `np.linalg.eig(A)` in Python.

**The following questions are on the singular value decomposition.**

(6) Suppose that $\mathbf{u}_1, \ldots, \mathbf{u}_n$ are orthonormal, and that $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are also orthonormal. Let $\sigma_1 > \sigma_2 > \ldots > \sigma_r > 0$. Show that $A_r = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ has rank $r$. *Hint: Find a basis for the range of $A_r$.*

**The following questions are on tensors.**

(7) Let $\mathcal{G}$ be a $2 \times 2 \times 2$ tensor with slices:

$$\mathcal{G}(:,:,1) = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix} \quad \mathcal{G}(:,:,2) = \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}$$

Furthermore, let:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad B = \begin{bmatrix} -2 & 2 \\ 3 & -3 \\ -4 & -4 \end{bmatrix} \quad \text{and } C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Calculate $\mathcal{G} \times_1 A \times_2 B \times_3 C$.